

Spin Logic — Comprehensive User & Installation Manual

Plan, Balance, and Cost Cotton Spinning Operations (Blow Room to Winding)

1. System Overview & Architecture

Spin Logic is an industrial-grade, offline-first calculation and planning suite engineered specifically for cotton spinning mills. Built as a unified Flutter codebase, it provides real-time production calculations, raw material blend feasibility, ring frame doff timing, complete departmental balance, and full mill financial accounting from blow room to autoconer winding.

Key Technical Characteristics:

- **Dual Target Platform:** Runs natively as an Android App (APK/AAB) and as a zero-latency responsive Web Application.
- **100% Offline Capable:** Calculations, unit conversions, and session data operate completely without internet connectivity.
- **Live Calculation Engine:** Outputs update continuously as values are typed — no recalculate buttons required.
- **Enterprise Top Bar:** Standardized actions on all module pages: *Back to Home*, *Save data*, *Print*, and *Save PDF*.

2. Module-by-Module Functional Guide

Module 1: Production Calculation

Departmental production calculator covering Carding, Draw Frame, Comber, Simplex (Speed Frame), Ring Frame, and Winding (Autoconer).

- **Dynamic Department Switcher:** Modifies all input parameters and reference constants based on selected department.
- **Ring Frame Spinning:** Computes TPI, OPS (oz/spindle/shift), Lbs/shift, Kg/shift, and Bags/day using exact mill twist and spindle speed formulas.
- **Carding & Drawing:** Models delivery speed (m/min), sliver hank / grains/yard, and machine efficiencies to output sliver weights and bags/day.
- **Comber:** Computes production based on nips/min, feed length (mm), noil extraction %, heads per machine, and lap grains/yard.
- **Simplex:** Calculates delivery rates, roving hank, flyer RPM, TPI, and flyer package outputs.
- **Winding:** Computes autoconer drum production at speeds up to 1400 m/min, efficiency, and winding spindles required.

Module 2: Count Conversion & Linear Density

Universal textile count conversion engine and sample linear density analyzer.

- **All Major Systems Supported:** Indirect systems (Ne English Cotton, Nm, Worsted Nw, Linen Lea, Woollen Yorkshire) and Direct systems (Tex, Denier, Dtex, Ktex, Grains/yd).

- **Unified Tex Pivot:** Mathematically rigorous bidirectional conversions: $Ne = 590.5/Tex$, $Nm = 1000/Tex$, $Denier = Tex \times 9$, $Dtex = Tex \times 10$.
- **Linear Density Calculator:** Computes yarn count directly from sample length and weight. Includes one-touch presets for Lea (120 yd), Hank (840 yd), 100m, 10m, 1m, and 1 yd with full unit conversions (g, mg, kg, grain, oz, lb).

Module 3: Blow Room & Card Waste

Comprehensive waste analysis and combined process loss auditor.

- **Cotton Mixing Table:** Multi-component cotton blend matrix with individual yield and moisture percentages.
- **Non-Usable Wastages:** Tracks droppings, sweepings, hard waste, AC fan extraction, and invisible moisture loss.
- **Department Measured Yields:** Feed vs. waste inputs on a kg basis for Blow Room lines and Cards.
- **Combined Yield Calculation:** Computes compound blow room → card yield and net process waste.

Module 4: Profit/Loss Calculations

Comprehensive cotton blend costing and mill profitability engine across 5 integrated sections.

- **1. Raw Material Cost:** Maund to Kg conversions (37.3242 standard), Rate/Kg to Rate/Maund, Carded (CD) vs. Combed (CM) yields, Comber noil deductions, and cost/lb.
- **2. Net Raw Material Cost:** Evaluates comber noil market value credit to calculate net clean combed fiber costs.
- **3. Making Charges:** Spindle-cost making charges per shift/day, frame allocation, and per-lb manufacturing conversion cost.
- **4. Count-Wise P&L:** Combines raw material costs, making charges, and channel packing rates (Export vs. Local) against sale prices to compute net margin/lb.
- **5. Mill Summary:** Projected bottom-line mill profit or loss per Day, per Month (working days basis), and per Year.

Module 5: Pressure Conversion

Precision pneumatic converter for mill utility systems.

- **Utility Coverage:** Covers compressor lines, suction fans, blow room transport lines, and spinning nozzle pressures.
- **Units & Decimals:** Simultaneous output for Bar (4 dec), PSI (3 dec), kg/cm^2 (4 dec), kPa (2 dec), mmHg (1 dec), Atm (4 dec), and mBar.

Module 6: Relative Humidity (Psychrometer)

Calculates air humidity from wet & dry bulb readings using psychrometric equations.

- **Psychrometer Calibration:** Select between Aspirated / Sling psychrometers ($A=0.000662$) and Natural Draft psychrometers ($A=0.0008$).
- **Atmospheric Compensation:** Pressure-compensated vapor pressure calculations.
- **Mill Standards Guidance:** Built-in reference ranges for Blow Room / Carding (55–60%), Ring Spinning (45–55%), and Winding (60–70%).

Module 7: Target Count Feasibility

HVI fiber property screening against target yarn count (Ne).

- **Input Screening:** Evaluates fiber length (mm), uniformity index (%), micronaire ($\mu g/in$), strength (g/tex), short fiber content (%), and maturity.
- **Cross-Section Analysis:** Computes fibers per yarn cross-section (Yarn Tex / Fiber Tex) and target TPI.

- **Risk & Trial Guidelines:** Evaluates spinnability safety thresholds and flags critical limiting factors (e.g., low fiber cross-section count or high short fiber content).

Module 8: Ring Doff & Roving Consumption

Production timing and roving package logistics calculator.

- **Auto vs. Manual Modes:** Auto mode linearly interpolates spindle speed, TPI, bobbin weight, and roving package weight from 20s to 80.2s reference tables.
- **Ring Doff Timing:** Calculates runtime per doff change and total doff cycles per 24-hour day.
- **Roving Logistics:** Computes roving packages consumed per frame per day, per shift, and per doff cycle, plus package runtime hours.

Module 9: New Mills Plan

End-to-end mill capacity planner from market bag demand to front-line machinery.

- **Department Machine Sizing:** Sizes required Blow Room lines, Cards, Breaker drawing, Combers, Lap formers, Finisher drawing, Simplex flyers, Ring frames, and Autoconer spindles.
- **Speed Balancing Modes:** Toggle between 'Reducing speed (whole machines)' to align actual production to target bags or 'Decimal machines (full speed)'.

Module 10: Spin Plan (Auto-Balance)

Departmental machine speed and count allocation balancing engine.

- **Ring-to-Blow Room Flow:** Backwards balanced flow with waste allowances across every process stage.
- **Fleet Balancing:** Evaluates available fleet vs. required fleet and computes required operational run speeds (% of standard rated speed).
- **Status Indicators:** Live shortfall alerts when department capacity cannot meet production targets.

3. App Installation & Deployment Guide

A. Web Deployment (Vercel)

The web application builds as a production-optimized Single Page Application (SPA). The project is pre-configured with `vercel.json` rewrites.

1. **Build Command:** Run `flutter build web --release` in the project root directory.
2. **Output Directory:** The generated web distribution is located in `build/web`.
3. **Vercel CLI Deployment:** From the repository root, execute `vercel --prod` or connect your GitHub repository to Vercel.
4. **Vercel Project Settings:** Framework Preset: *Other*, Output Directory: *build/web*, Build Command: *flutter build web --release*.

B. Android Mobile App Installation (APK & AAB)

The Android version provides mill managers and technicians on the factory floor with a native, high-performance touch interface that works without internet connectivity.

1. **Building APK (Direct Installation):** Execute `flutter build apk --release`. The installer APK will be generated at `build/app/outputs/flutter-apk/app-release.apk`.
2. **Sideload on Android Phones:** Transfer `app-release.apk` to your device via USB cable, Google Drive, or WhatsApp. Tap the APK file and select 'Install' (ensure 'Install Unknown Apps' is permitted in Android Settings).
3. **Building App Bundle (Google Play Store):** Run `flutter build appbundle --release` to produce the Google Play publishing bundle at `build/app/outputs/bundle/release/app-release.aab`.
4. **Offline Cache:** All reference tables and calculating routines run purely in client memory — no network permissions are required for core calculations.

C. Developer & Desktop Testing

- **Local Web Server:** Run `flutter run -d chrome` for interactive development.
- **Automated Test Suite:** Execute `flutter test` to verify calculation accuracy across all pure Dart logic modules.
- **Static Analysis:** Run `flutter analyze` to guarantee strict null-safety and lint compliance.

4. Customizing Mill Reference Data

All benchmark tables (spindle speeds, twist multipliers, efficiencies, cotton spinnability ranges) are externalized in editable JSON files located in `assets/data/`:

- `ring_doff_reference.json`: Reference speeds, TPI, bobbin & roving weights for Ne 20 to 80.2.
- `production_reference.json`: Machine efficiencies, twist multipliers, and departmental delivery speeds.
- `count_potential.json`: Fiber length and micronaire screening matrices.

Mill managers can modify these tables without altering Dart source code.